GENERATIVE AI & DOCUMENT INTELLIGENCE

StudyLens AI

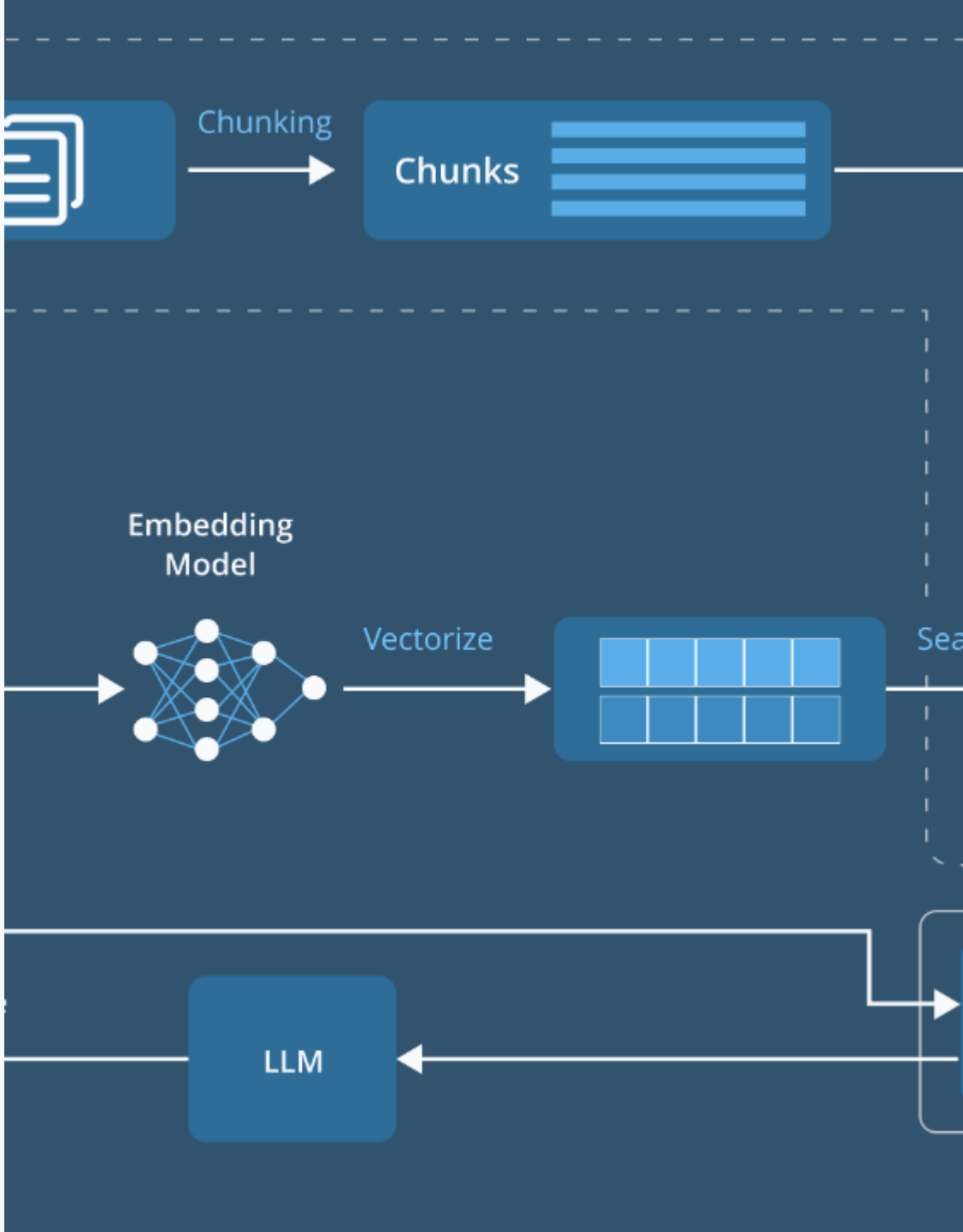
Transforming static documents into interactive knowledge graphs using RAG.

github.com/tmthangaraj1980-AIML/Studylense_AI

THE PROBLEM

Information Silos

- **Data Overload:** Massive volume of unstructured enterprise PDFs.
- **Retrieval Friction:** Keyword search fails contextual intent.
- **Static Context:** Inability to cross-reference multiple dense sources.



Core Objectives

Semantic Ingestion

Parse complex layouts accurately.

Instant Retrieval

Sub-second vector similarity search.

Grounded Generation

Zero-hallucination factual synthesis.

Technology Stack



Python 3.10



FastAPI



LangChain



ChromaDB

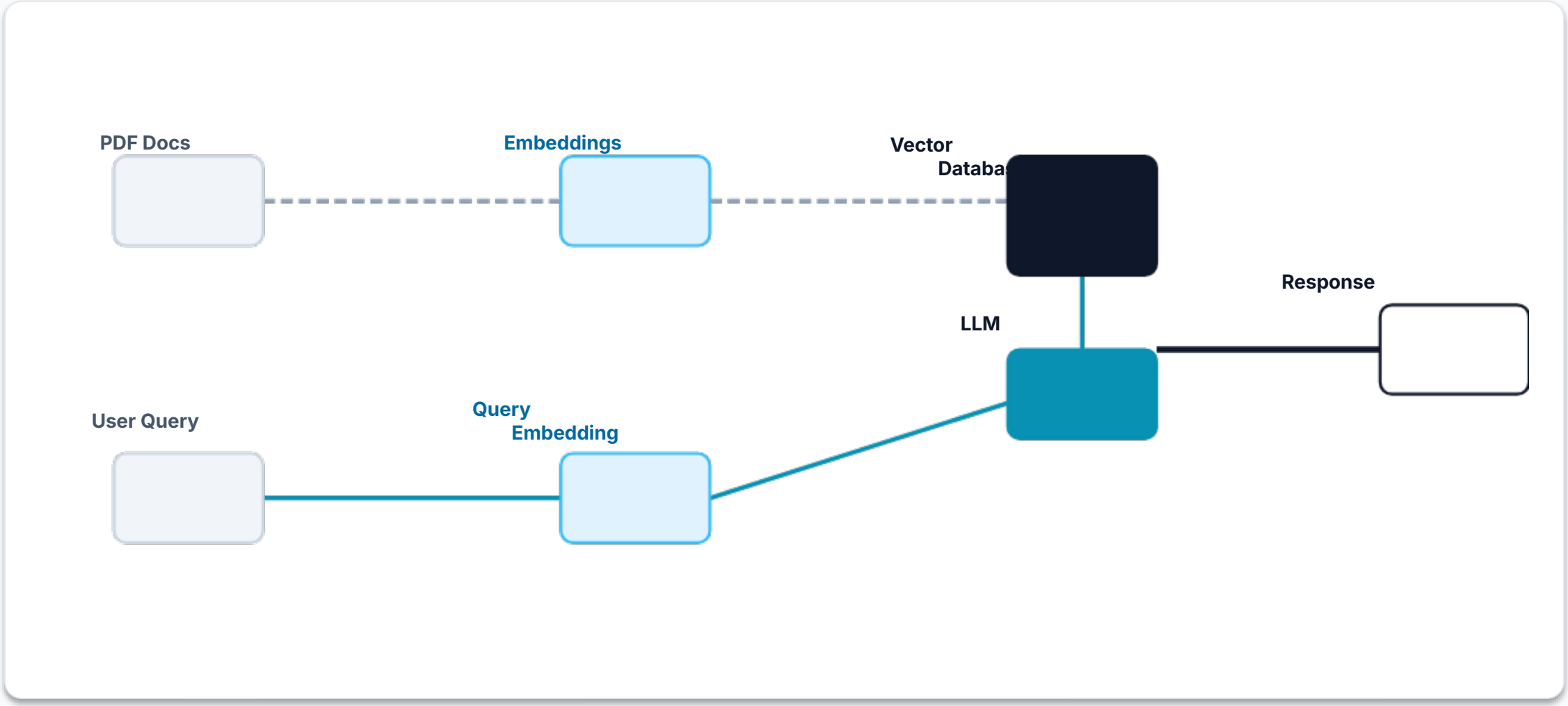


OpenAI / LLMs

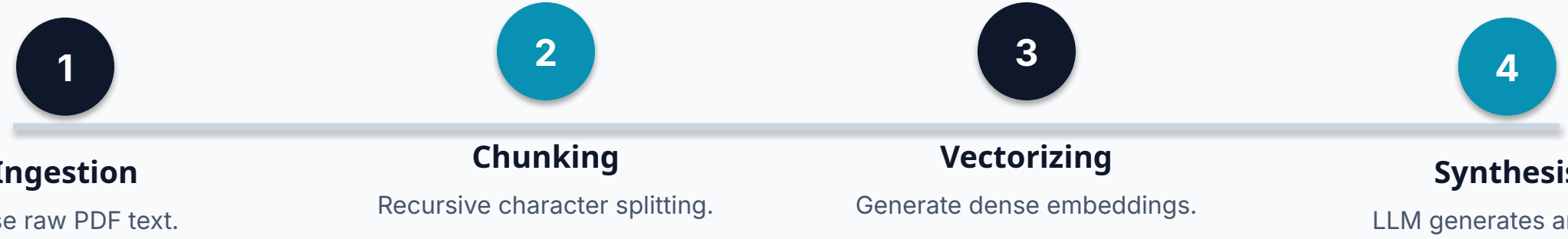


PyPDF2

RAG Architecture



Processing Pipeline



Dataset & Preprocessing

Source Material

- Academic research papers.
- Corporate technical manuals.
- Unstructured text corpora.

Cleaning Pipeline

- Whitespace normalization.
- Non-ASCII character removal.
- Semantic chunking strategies.

Core AI Algorithms

Dual-model architecture powering the intelligence engine.

Embedding Model

Model: OpenAI text-embedding-ada-002

Function: Maps text chunks to 1536-dimensional vectors.

Generation LLM

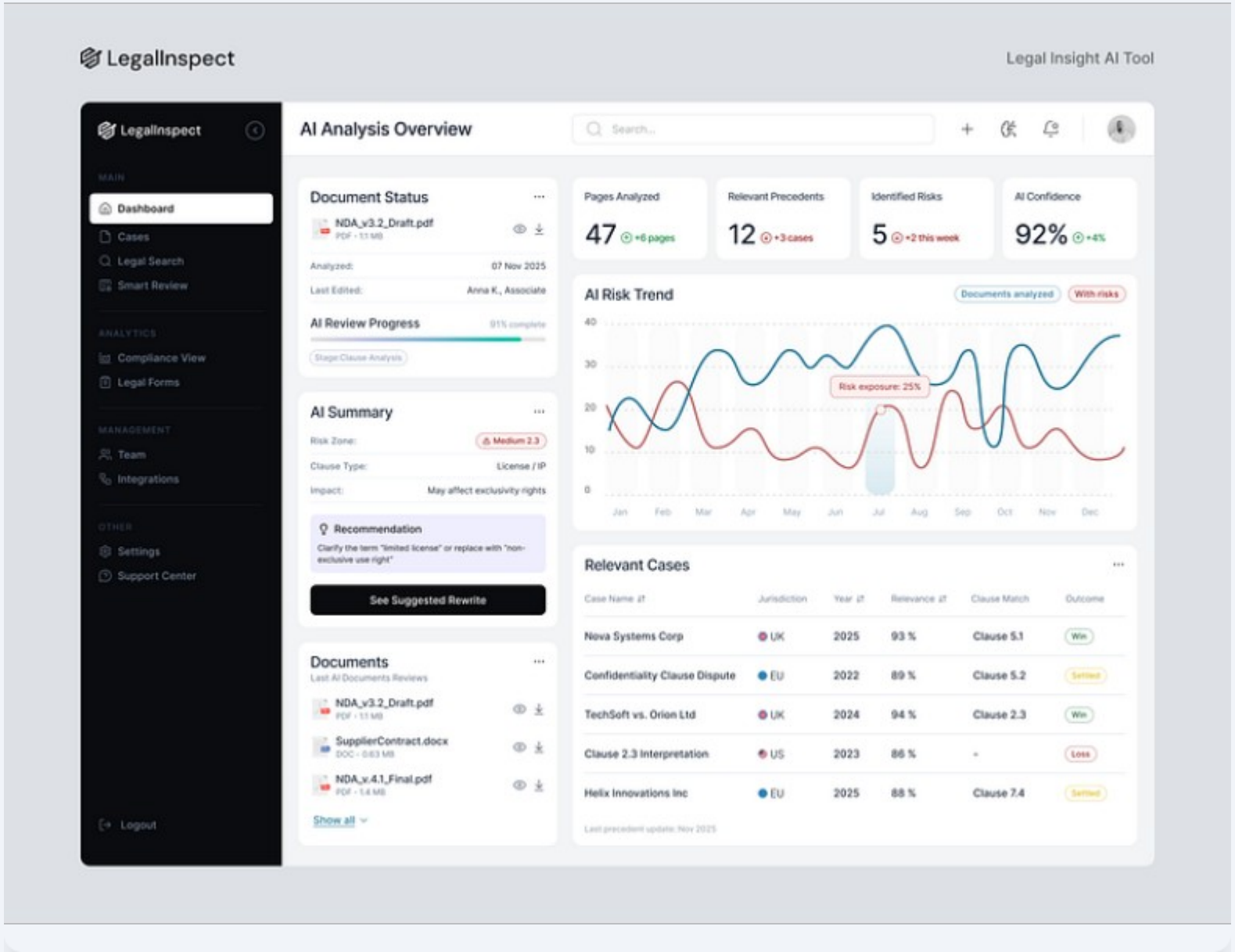
Model: GPT-3.5-Turbo / GPT-4

Function: Synthesizes answers bounded by retrieved context.

Application Interface

Clean, intuitive chat interface facilitating real-time document interrogation.

- **Live Status: Deployed**



Challenges & Iterations

Encountered Challenges

Context Window Limits Truncated large document retrieval results.

Table Extraction Loss of structural format in standard PDF parsers.

Future Improvements

Multimodal RAG Process images and diagrams alongside text.

Hybrid Search Combine keyword (BM25) with vector search.

Technical Q&A

1. How does RAG mitigate hallucination?

Grounds the LLM generation explicitly in retrieved factual context.

2. Why semantic over lexical search?

Captures contextual meaning rather than exact keyword matches.

3. Ideal chunking strategy?

Recursive character splitting with overlapping windows.

4. How to optimize Vector DB speed?

Utilize HNSW (Hierarchical Navigable Small World) indexing.

5. Handling massive documents?

Implement map-reduce summarization prior to querying.

Intelligent Insights Delivered.

StudyLens AI bridges the gap between static enterprise data and actionable knowledge extraction.



/Studylense_AI



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Image Sources



https://d3lkc3n5th01x7.cloudfront.net/wp-content/uploads/2024/08/26051537/Advanced-RAG.png

Source: www.leewayhertz.com



https://cdn.dribbble.com/userupload/45788503/file/b1676f6d71c218be5825d216432d1398.png?resize=752x&vertical=center

Source: dribbble.com